

The Best Speech-to-Text Engine for AI Voice Agents on the Telephony Layer

A Controlled Comparison of Five Streaming STT Engines
in One Identical PSTN Pipeline

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Abstract

On the telephony layer, the latency of a speech-to-text (STT) engine is a necessary but *insufficient* property: the objective that actually matters for an AI voice agent is **accuracy and reliability under real public switched telephone network (PSTN) conditions**. We swapped five streaming STT engines into one identical Pipecat telephony pipeline, holding turn-taking, language model, text-to-speech, and the 8 kHz audio path constant, and placed roughly ten live Twilio phone calls per engine (about fifty total), each requiring the assistant to capture a spoken name and a ten-digit callback number. We find that speed does not predict accuracy: the fastest-median engine (Cartesia Ink-Whisper) was the *least* reliable, hallucinating nine invented tokens and exhibiting a broken two-second latency tail. We also find that one does not have to trade speed for accuracy. We recommend the stack **Deepgram Nova-3 for STT paired with ElevenLabs Flash v2.5 for TTS** as the primary choice for telephony voice agents. Deepgram’s STT ties Speechmatics for the best first-try number accuracy (70%) with zero permanent number losses and zero hallucination, yet runs at roughly half the latency of Speechmatics (STT p50 0.374s vs. 0.777s), and it captured the hard name “Mazin” once (fewer than AssemblyAI’s three, but unlike Speechmatics’ zero), so it delivers accuracy, reliability, and speed in a single engine with no critical reliability weakness, though AssemblyAI captures the hard name more often. We pair it with ElevenLabs for synthesis because Deepgram’s own Aura text-to-speech, not its STT, interrupted and cut off speech on the telephony line; this also matches our production deployment. **AssemblyAI Universal-3 Pro** is the runner-up and leads proper-noun accuracy (three exact “Mazin,” the only engine to get the brand acronym), while **Speechmatics Ursa (Enhanced)** is an accuracy-focused alternative that ties on numbers (70%) with the cleanest digit formatting but runs $\sim 2\times$ slower with no name-capture advantage.

1 Introduction

Telephony is the hardest deployment surface for automatic speech recognition, and an AI voice agent that answers phone calls lives or dies on that surface. Four properties make it hard.

Narrowband audio. The PSTN delivers 8 kHz mu-law audio. Half the spectral information that wideband models are trained on is simply absent. No amount of upsampling restores the missing high frequencies; it only changes the container. An engine that is excellent on clean 16 kHz microphone audio can degrade sharply on a real phone call.

Names and numbers are the hard tokens. A receptionist’s core job is to capture *entities*: a caller’s name and a callback number. These are exactly the tokens with the least linguistic redundancy. A language model can repair a garbled common word from context; it cannot infer that “016” was meant when the line delivered “015”. Proper nouns are worse still, an uncommon name like “Mazin” has essentially no language-model prior to lean on, so the transcript is whatever the acoustic model emits.

Latency is felt every turn. In a spoken conversation, every turn pays the STT’s time-to-first-token (TTFB). Unlike batch transcription, where total wall-clock time is amortized, conversational latency is experienced repeatedly and directly as the gap before the agent responds. This makes TTFB a first-class metric, but, as we show, not the *only* one.

Hallucination is a safety issue. Some streaming models, when handed unparseable narrow-band audio, do not stay silent: they invent plausible tokens. For a transcription product this is a quality nuisance. For a voice agent acting on what it “heard,” a hallucinated phrase or a fabricated digit run is a correctness and trust failure, the agent confidently stores or acts on something the caller never said.

This paper argues a single thesis and supports it with controlled measurements: **on the telephony layer, optimize for accuracy and reliability, not raw speed; and you do not have to trade one for the other.** Speed alone does not imply accuracy, the fastest engine here is the least reliable, yet the best-balanced STT is also fast, so a slower engine buys no accuracy advantage. Reliability also spans the whole stack, not the STT alone: the agent must never hallucinate, must capture names and numbers correctly on a noisy 8 kHz line, and must not cut the caller off, a turn-taking property that, as we found, lives in the text-to-speech layer as much as anywhere else.

2 Experimental Setup

2.1 The constant pipeline

We built one Pipecat telephony pipeline and changed exactly one component, the STT engine, across all conditions. Everything else was held fixed:

- **Turn-taking:** Smart Turn V3 end-of-turn detection over Silero voice-activity detection (VAD), identical for every engine.
- **Language model:** Claude Haiku 4.5.
- **Text-to-speech:** ElevenLabs Flash v2.5.
- **Audio path:** Twilio PSTN at 8 kHz mu-law, upsampled *centrally* to a 16 kHz container (`audio_in_sample_rate` = 16000) before the STT, so every engine received bit-identical input framing. TTS output was rendered at 24 kHz.

2.2 Fairness controls

Three controls protect the comparison. First, **no vocabulary or keyterm seeding** was applied to any engine, no custom dictionaries, no boosted phrases, no biasing toward the test name, number, or brand. This is the honest out-of-the-box condition and it makes the hard-token results comparable. Second, **central upsampling** guarantees identical audio framing rather than letting each SDK resample differently. Third, **ground-truth integrity**: each batch’s STT processor class matched its label, so no engine’s output could be attributed to another (no cross-contamination of transcripts).

2.3 Engines and configuration

Table 1 lists the five engines and their exact streaming configuration. All are run unseeded.

Engine	Model / endpoint	Streaming configuration
Deepgram	Nova-3 (nova-3-general)	interim results; smart_format; punctuate; endpointing 500 ms
ElevenLabs	Scribe v2 Realtime	scribe_v2_realtime ; manual commit
Speechmatics	Ursa, ENHANCED	turn_detection_mode=EXTERNAL ; include_partials; max_delay = server default
AssemblyAI	Universal-3 Pro Streaming	u3-rt-pro ; vad_force_turn_endpoint=True
Cartesia	Ink-Whisper (ink-whisper)	default streaming

Table 1: The five streaming STT engines and their configuration. Only this component varied; all engines were run without vocabulary or keyterm seeding.

2.4 Instrumentation and the read-back accuracy proxy

We instrument STT TTFB per turn and report its mean, median (p50), and 95th percentile (p95). We also record the per-component medians for the LLM and TTS so that STT latency can be placed in the context of the whole turn.

Because we deliberately ran *without* ground-truth WER tooling, we measure entity accuracy with an **AI read-back proxy**: the assistant repeats the captured number back to the caller, and we score whether the spoken read-back matches the true number. The speaker said the same ten-digit number, 555-016-2049, clearly on essentially every call, so a mismatch is attributable to STT error rather than caller variation. The one exception is a single AssemblyAI call in which the caller volunteered a *different* number than the standard one; because that call does not test the engine against the fixed target, it is excluded from the integrity premise and flagged where it affects scoring (Section 3.2). We score four outcomes per call: correct on the *first try*; *fixed on retry* (the read-back-and-confirm loop repaired it); *never correct* (a permanently wrong stored number); and *eventual success* (correct by call end). Name and brand accuracy are scored as exact-match captures of the target proper noun “Mazin” and the rare phrase “MM Intelligence.”

2.5 Sample

One single speaker placed roughly ten live Twilio (PSTN) calls per engine, about fifty calls in total. The task on every call was identical: book an appointment and provide a name and a ten-digit callback number. This sample is small and single-speaker by design, enough to expose decisive reliability differences, and we are explicit about its limits in Section 5.

3 Results

3.1 Latency

Table 2 reports STT TTFB and the per-component medians. The headline is that, *end to end*, the engines are close: the top four by total p50 fall within ~108 ms of one another (0.96–1.07 s). Speechmatics adds roughly 0.4–0.5 s per turn at the STT stage. Cartesia is **bimodal**: it has the fastest median (0.296 s) but a p95 of 2.049 s, choking for about two seconds on disfluent input.

Engine	STT mean	STT p50	STT p95	LLM p50	TTS p50	TOTAL p50
Deepgram	0.366	0.374	0.465	0.540	0.144	1.058
Scribe	0.340	0.345	0.393	0.587	0.139	1.071
Speechmatics	0.768	0.777	0.952	0.592	0.141	1.510
AssemblyAI	0.375	0.357	0.443	0.521	0.134	1.011
Cartesia	0.994	0.296	2.049	0.531	0.136	0.963

Table 2: Latency in seconds. STT time-to-first-token (mean, p50, p95) plus per-component medians. TOTAL p50 is the sum of medians (STT p50 + LLM p50 + TTS p50), computed from the unrounded medians; it is an approximate composite, *not* a measured end-to-end percentile, and excludes turn-detection wait and network time. (Because it uses unrounded inputs, the rounded AssemblyAI parts may not visibly re-add to its printed total.) Bold marks Cartesia’s bimodal extremes: fastest median, broken tail.

Figure 1 shows the TTFB p50/p95 bars. Cartesia’s p95 tail and Speechmatics’ higher-but-stable bars are both visible; the other three sit tightly under 0.47 s at p95. Figure 2 stacks the per-turn medians and makes the structural point: STT is a *minority* of the turn budget, and the engines are nearly indistinguishable end to end, so optimizing STT purely for speed buys little while risking the reliability axis.

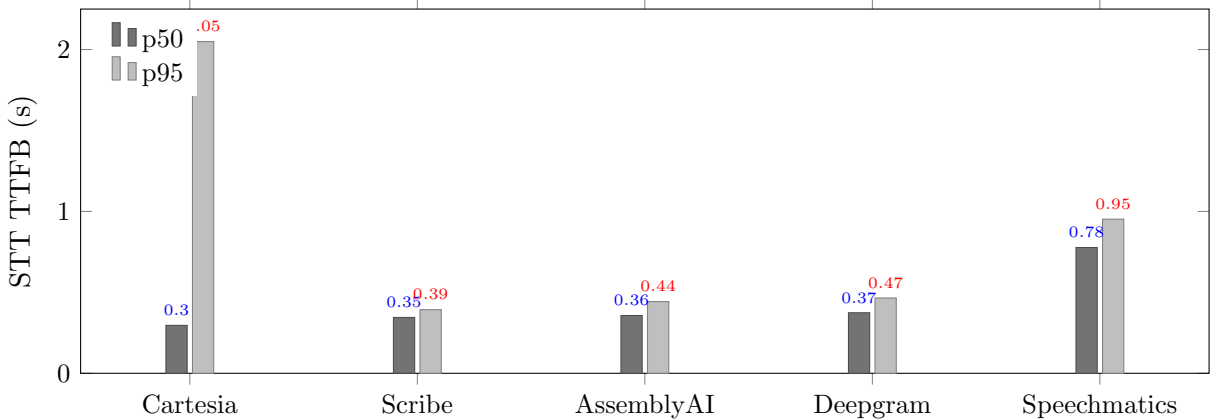


Figure 1: STT time-to-first-token, p50 vs. p95, per engine (ascending by p50). The two statistic series use neutral grays so that engine brand colors remain reserved for the per-engine charts; near-coordinate labels are shown to two decimals, with the exact three-decimal figures in Table 2. Cartesia owns the fastest median yet a 2.049 s p95 tail, a broken worst case. Speechmatics’ bars are the tallest but tight (p50 0.777 s, p95 0.952 s): slower, but stable and predictable.

3.2 Number accuracy

Table 3 reports read-back accuracy for the callback number. Speechmatics and Deepgram tie for the best first-try number accuracy (70%), both with zero permanent losses; Speechmatics retains a distinct edge in the cleanest auto-formatting ((555) 016-2049). The first-try gap among the top three (70/70/50%) spans only about one call and is within noise; we do not over-rank it.

The decisive column is **never**, a permanently wrong stored number, the failure a receptionist cannot walk back. Only AssemblyAI (1) and Cartesia (1) ever lost a number permanently; Speech-

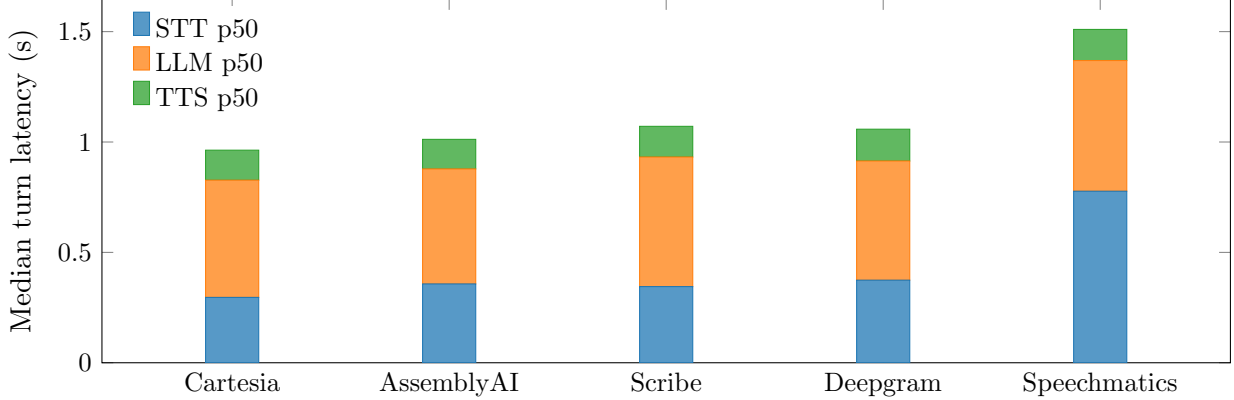


Figure 2: Per-turn latency budget, stacked medians (STT + LLM + TTS). STT is a minority of every turn. The four non-Speechmatics engines land within ~ 108 ms end to end (0.96–1.07 s); Speechmatics’ extra cost is confined to the STT segment. Composite of medians, not a measured end-to-end percentile.

matics, Deepgram, and Scribe lost none. The assistant’s read-back-and-confirm loop is what lifts every engine to 90–100% eventual accuracy, which is precisely why a single first-try number does not settle the ranking but a single permanent loss does.

Two of these counts rest on call-level judgment and we make them explicit. AssemblyAI’s lone permanent “loss” is a borderline case: on that call the caller actually volunteered a *different* number than the standard 555-016-2049 and the agent captured and confirmed it correctly, so it is not strictly an STT failure against the fixed target. We retain it in the “never” count conservatively (the agent did not end the call holding the standard number) but flag that excluding it would make AssemblyAI 4/9 first-try, never = 0, eventual 100%.¹ Cartesia’s permanent loss is the call in which a wrong number was affirmatively confirmed mid-conversation before a later correct read-back; a naive “final read-back” check would miss it and score Cartesia never = 0.²

Engine	First-try	Fixed on retry	Never	Eventual	Note
Speechmatics	7/10 (70%)		3	0	100% cleanest formatting
Deepgram	7/10 (70%)		3	0	100% ties best first-try
Scribe	5/10 (50%)		5	0	100%
AssemblyAI	4/10 (40%)		5	1	90% loss is a different-number call
Cartesia	2/10 (20%)		7	1	90% loss confirmed mid-call

Table 3: Callback-number read-back accuracy (standard number 555-016-2049, spoken identically on every call except one AssemblyAI call where the caller gave a different number). “Never” counts permanently wrong stored numbers and is the decisive reliability column. Speechmatics and Deepgram tie for the best first-try accuracy (70%); see the footnotes in Section 3.2 for the two judgment-dependent “never” counts.

Figure 3 is the thesis figure. It plots STT p50 latency against first-try number accuracy. There

¹This is the single call that motivates the softened integrity premise in Section 2: the standard number was *not* spoken on it, so it does not test the engine against the constant target.

²The Cartesia read-back transcript for this call is non-monotonic in time: a wrong number (“...five five nine, zero one, one four one”) is confirmed before a later correct read-back of 5550162049 appears. The permanent loss is counted because a wrong number was affirmatively confirmed, not because the call ends incorrectly.

is **no positive correlation between speed and accuracy**: Cartesia is the fastest yet least accurate (0.296 s, 20%); Speechmatics is the slowest yet most accurate (0.777 s, 70%). We draw no trend line because there is no trend to draw. Figure 4 contrasts first-try and eventual number accuracy and shows how the confirm loop compresses the field at the eventual stage, while leaving the two engines with permanent losses capped at 90%.

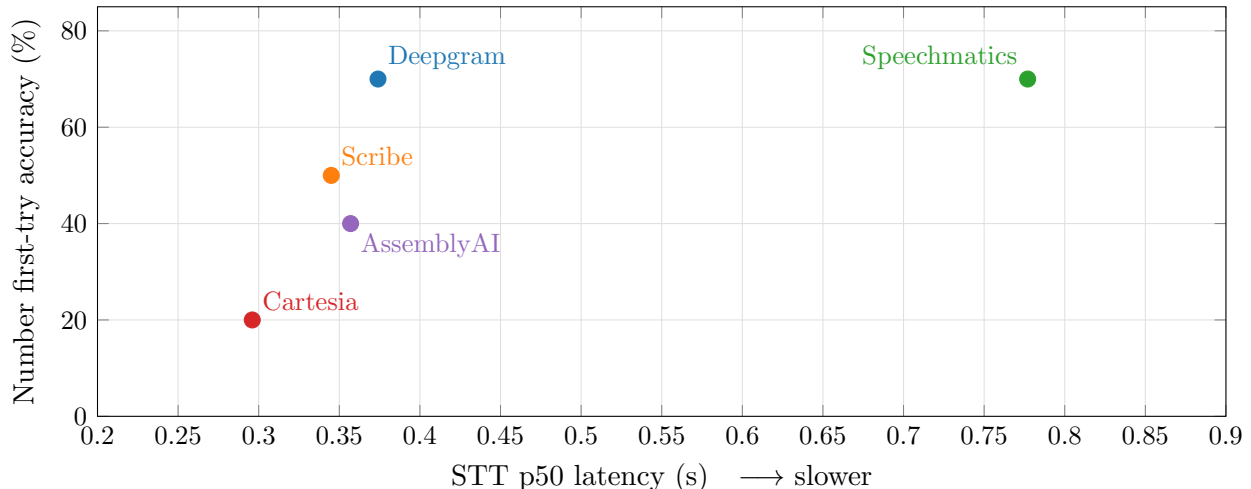


Figure 3: The thesis figure: STT p50 latency vs. first-try number accuracy. One labeled point per engine. Fastest $\neq$ most accurate, Cartesia is fastest yet least accurate (bottom-left), while Speechmatics and Deepgram tie for the best first-try accuracy (70%) at opposite ends of the latency axis (Speechmatics slowest, top-right; Deepgram far faster at the same height). There is no positive speed–accuracy correlation, so no trend line is drawn.

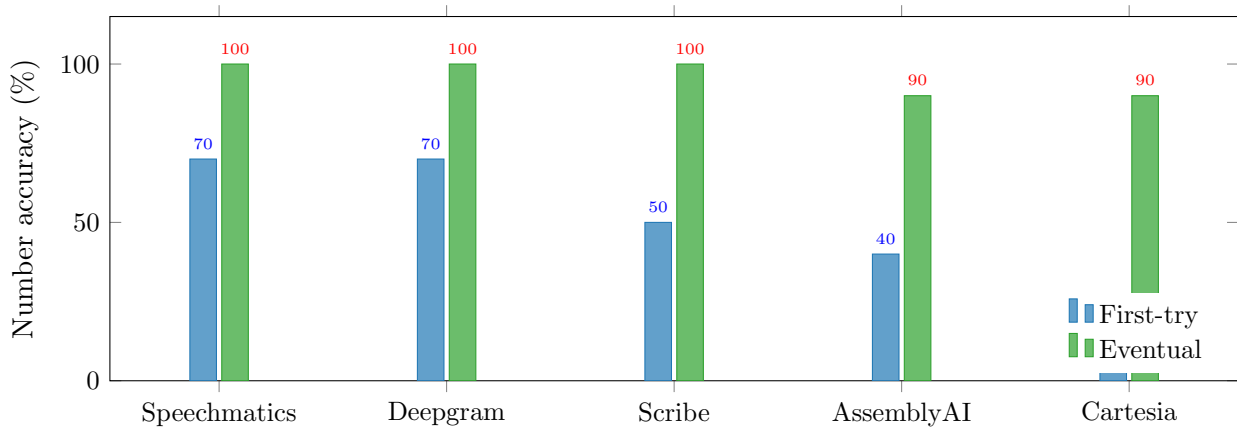


Figure 4: Number accuracy: first-try vs. eventual, per engine. The read-back-and-confirm loop lifts every engine to 90–100% eventual, but AssemblyAI and Cartesia are capped at 90% by their single permanent losses, while the three loss-free engines reach 100%.

3.3 Name accuracy

Proper nouns separate the engines where digits compress them. Table 4 reports exact captures of the target name “Mazin” and lists the raw `caller_names` each engine produced. AssemblyAI is

the clear leader with three exact “Mazin” captures; Scribe, Cartesia, and Deepgram manage one each; only Speechmatics never captured it exactly, returning near-misses such as “Maison” and “Mohsen.” Common names (Emma, Tommy, Michelle) were handled well by all engines and are non-discriminating at this sample size.

Engine	Exact “Mazin”	Captured caller_names (raw)
AssemblyAI	3	Mazin ×3, Emma, Tommy, Michelle, Nabi
Scribe	1	Mazin, Martin, Neha, Navy, Jack, Tommy, Josh, Michelle, Emma, Nicolas Cage
Cartesia	1	Mazin, Mazza, Mazen, Mazen Saleem, Basen, Emma, John, Tom
Deepgram	1	Mazin, Maazel, Mazan Salim, Maazen, Austin, Tommy, Michelle, Neha, Salim, Bright
Speechmatics	0	Maison, Maison, James, Emma, Mohsen, Mohsen, Jack, Vanessa, Hannah

Table 4: Exact captures of the hard proper noun “Mazin” (the speaker’s real name) and the raw captured names per engine, unseeded. AssemblyAI leads entity capture; Deepgram lands the hard name once, while Speechmatics misses it entirely.

3.4 Brand / rare phrase

Table 5 reports how each engine rendered the rare phrase “MM Intelligence”, specifically the word before “intelligence.” Only AssemblyAI ever produced the correct “mm” acronym (and only once, among three “mm” renderings). Every other engine substituted plausible but wrong tokens (“ml,” “of,” “amazon,” “artificial,” “animal”). Unseeded, a novel brand acronym is near-impossible for narrowband STT, further evidence that entity capture, not raw speed, is the binding constraint.

Engine	Rendering of the word before “intelligence”
AssemblyAI	mm ×3 (only one correct)
Cartesia	mm ×1
Deepgram	of / mim
Scribe	mm / artificial / amazon / animal ×2 / m
Speechmatics	ml ×2 / mml / with ×3 / m

Table 5: Rendering of the rare brand phrase “MM Intelligence” (target word: “MM”). Only AssemblyAI produced a correct “mm”; all engines otherwise substituted. “mm” captures are counted leniently, they include hyphenated/split forms such as “M-M” and “Emma M.”, and “only one correct” for AssemblyAI is a manual case/spacing judgment not derivable from the lowercased transcript text alone. Unseeded, novel acronyms are a known hard case.

3.5 Hallucination

Table 6 is the sharpest reliability separator. We counted invented “whatever” tokens, content the model emitted for unparseable narrowband audio that the caller never said. **Cartesia produced nine**; every other engine produced *zero*. Cartesia additionally emitted repeated-digit runs (e.g. “016-016-016,” “0-1-6-9-9-9”), the Whisper failure mode of filling silence or noise with fabricated structure. For a voice agent this is disqualifying: the agent acts on tokens that were never spoken.

Engine	Invented tokens	Other artifacts
Cartesia	9	repeated-digit runs (016-016-016; 0-1-6-9-9-9-9)
Deepgram	0	,
Scribe	0	,
Speechmatics	0	,
AssemblyAI	0	,

Table 6: Hallucinated (“whatever”) tokens invented in caller turns. Cartesia fabricated nine and emitted repeated-digit runs; all other engines hallucinated nothing.

3.6 Reliability summary

Table 7 consolidates the failure modes that actually break a receptionist, and Figure 5 visualizes the three decisive axes together. Cartesia’s hallucination spike and broken latency tail, and AssemblyAI’s name-capture lead, are both immediately visible. Deepgram is clean on every measured STT reliability axis (zero permanent losses, zero hallucination, one hard-name capture) while also posting a sub-second, stable tail, the only engine to combine top-tier accuracy with low latency.

Failure mode	Speechmatics	AssemblyAI	Deepgram	Scribe	Cartesia
Permanent number loss	0	1	0	0	1
Hallucination (tokens)	0	0	0	0	9
Latency tail p95 (s)	0.952	0.443	0.465	0.393	2.049
Exact name captures	0	3	1	1	1

Table 7: Reliability summary across the failure modes that break a phone receptionist (all STT failure modes). The two permanent number losses are the judgment-dependent calls footnoted in Section 3.2. Bold in the latency-tail row marks Deepgram’s recommended best-balanced profile, not the row minimum (Scribe’s 0.393s p95 is the lowest in that row). Deepgram has zero permanent losses, zero hallucination, a hard-name capture (though AssemblyAI leads name capture 3 to 1), and a sub-second, stable tail, the best balance of accuracy and speed; Speechmatics matches it on the two failure axes but at $\sim 2\times$ the latency. The caller-cut-off behavior we observed was a property of the text-to-speech layer, not the STT (Section 4), so it is not an STT failure mode here.

3.7 Ground-truth integrity

Two facts protect the integrity of these numbers. The callback number was the *same* on essentially every call, the one documented exception is the single AssemblyAI call in which the caller volunteered a different number (Section 3.2), so number misses are STT errors, not caller variation. And each batch’s STT processor class matched its label, so no transcript from one engine was attributed to another. We do not claim a ground-truth word error rate; the entity metrics above are exact-match scores against known targets, which is the relevant unit for a receptionist anyway, a name or number is either right or it is not.

4 Discussion

Optimize for accuracy and reliability, and you need not trade away speed. The latency results (Section 3.1) and the thesis figure (Figure 3) make the central claim concrete. Speed does

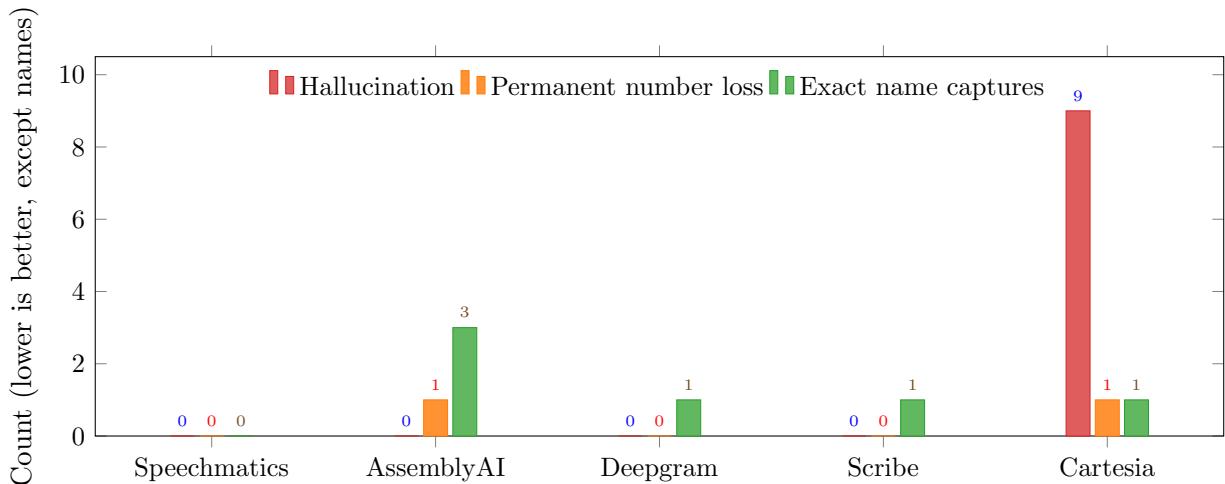


Figure 5: Reliability failure modes per engine. Hallucination and permanent number loss are failures (lower is better); exact name captures are a benefit (higher is better). Cartesia’s hallucination spike (9) dominates the chart; AssemblyAI leads exact name captures (3); Speechmatics shows no failures on either failure axis.

not imply accuracy: the fastest-median engine, Cartesia, was the least reliable on every accuracy and safety axis we measured. But the inverse trap is just as real, and the data refutes it. You do *not* have to buy accuracy with latency. Deepgram delivers accuracy, reliability, *and* speed in one engine: it ties Speechmatics for the best first-try numbers (70%) with zero permanent losses and zero hallucination, lands the hard name once, and does it at roughly half Speechmatics’ STT latency (p50 0.374s vs. 0.777s). Speechmatics’ extra $\sim 2\times$ at the STT stage therefore buys no accuracy advantage over Deepgram. Because STT is a minority of the turn budget (Figure 2) and the field is within ~ 108 ms end to end, shaving STT milliseconds at the cost of reliability is a bad trade, but so is paying latency for accuracy you can already get faster. Latency is a floor requirement, not the optimization target, and here the most accurate STT is also one of the fastest.

For a voice agent, performance *is* reliability, across the whole stack. The properties that decide whether a phone agent is usable are not captured by a single accuracy headline. They are: never hallucinating, capturing names and numbers correctly on a noisy 8 kHz line, and never cutting the caller off. Reframed this way, “the best STT” is the one that fails the least in the ways that break a receptionist, which is exactly how Table 7 ranks them. But reliability is a property of the *stack*, not the STT in isolation: as we discuss below, the caller-cut-off behavior we observed came from the text-to-speech engine, not the recognizer, so the reliable configuration is a reliable STT paired with a reliable TTS.

The best-balanced STT: Deepgram Nova-3. Deepgram’s STT is the best-balanced engine in the study: no critical reliability weakness, and competitive on both entity axes. Its numbers are top-tier: 70% first-try number accuracy, which *ties Speechmatics for the best in the field*, zero permanent losses, and zero hallucination. It also lands the hard name “Mazin” once, where Speechmatics never did, so it avoids the zero on both entity axes, solid on numbers and at least non-zero on the hard name, though AssemblyAI captures that name more often (three vs. one). And it does all of this fast: STT p50 0.374s, roughly half Speechmatics’ 0.777s, with a tight 0.465s p95. It is the only

engine that combines best-in-class number accuracy, zero hallucination, a hard-name capture, and low, stable latency at once. That is why we make Deepgram STT the primary recommendation, and pair it with ElevenLabs for synthesis (next paragraph), which is also the configuration we run in production.

Reliability spans the stack: the cut-off behavior was the TTS, not the STT. Reliability on telephony has an axis that isolated STT transcription metrics do not see: whether the agent cuts the caller off. In our telephony testing we observed exactly that failure, the agent interrupting and clipping speech, but on tracing it the cause was the *text-to-speech* layer, not the recognizer. Deepgram’s own Aura TTS exhibited interruption and speech-cut-off behavior on the line, which, together with lower output quality, is why we synthesize with ElevenLabs Flash v2.5 rather than Aura. We present this as a *qualitative, observed* field finding that motivated the TTS choice. The present benchmark held TTS constant at ElevenLabs across all five STT conditions and therefore did *not* measure Aura’s interruptions; we did not instrument a numeric interruption rate for Aura, and we do not invent one. Quantifying TTS barge-in behavior under controlled conditions is future work. The lesson is that strong STT metrics in isolation do not equal a reliable telephony agent: reliability is a property of the whole stack, and the reliable stack here is Deepgram STT paired with ElevenLabs TTS.

The names-vs-numbers trade-off, and why Deepgram resolves it. The cleanest mirror image runs between two engines. AssemblyAI led proper nouns (three exact “Mazin,” the only correct brand acronym) but trailed on first-try numbers (40%) and took one permanent number loss. Speechmatics inverted that profile exactly: best (tied) numbers and cleanest digit formatting, zero permanent losses, but the only engine with zero exact captures of the hard name. Deepgram sits between the two poles and, crucially, does not pay for it in latency: it ties Speechmatics for the best first-try numbers (70%, zero permanent losses) *and* lands the hard name once, so it avoids the zero on both entity axes at once while running at roughly half Speechmatics’ STT latency. That is why Deepgram is the best-balanced default rather than a niche pick. The two remaining engines are specialists. Speechmatics is the **accuracy-focused alternative**: it matches Deepgram on numbers and has the cleanest digit auto-formatting, but it is $\sim 2\times$ slower and never captured “Mazin” exactly, so its added latency buys no accuracy advantage over Deepgram, recommend it only where digit formatting matters most and latency is irrelevant. AssemblyAI is the **best pick for names**: choose it when proper-noun fidelity dominates and the weaker first-try number accuracy (40%, one permanent loss) is acceptable.

Fixable vs. unfixable errors, and the case for seeding. A first-try number miss is *fixable*: the read-back-and-confirm loop repaired enough of them to lift every engine to 90–100% eventual accuracy (Figure 4). A permanent loss and a hallucination are *unfixable* at the conversation layer, the agent stores or acts on wrong content without ever knowing. This distinction is why we weight the “never” and hallucination columns far above first-try differences of one call. It also points at the most obvious next lever: we ran every engine *unseeded*, and the worst residual errors were hard proper nouns and a novel brand acronym, exactly the tokens that vocabulary/keyterm seeding targets. Seeding the name, the brand, and digit-formatting hints should compress the name and brand gaps materially and is the first thing to add before any production deployment.

Cartesia is concrete proof that speed $\neq$ reliability. Cartesia had the fastest median TTFB and was nonetheless disqualifying on telephony: nine hallucinated tokens, fabricated repeated-digit

runs, a permanent number loss, the worst first-try number accuracy (20%), and a broken 2.049s p95 tail. It is the cleanest single counterexample to the assumption that the fastest STT is the best STT.

TTS note (design choice, not an STT result). Synthesis was held constant across all conditions with ElevenLabs Flash v2.5, so no number above depends on the TTS choice. We record the qualitative field observation that motivated that choice: Deepgram’s own Aura TTS exhibited interruption and speech-cut-off behavior on the telephony line, and its output quality was lower, while ElevenLabs Flash v2.5 was more natural and did not clip the caller. Because this benchmark held TTS fixed at ElevenLabs, it did *not* measure Aura’s interruptions; we report this as a qualitative observation, not a measured rate, and we do not assign Aura a numeric interruption rate. Quantifying TTS interruption behavior under controlled conditions is future work. This is a design judgment about the TTS component, not a measured STT result, and it does not affect any number above.

5 Conclusion

For AI voice agents on the telephony layer, the objective is accuracy and reliability under real PSTN conditions, and the data shows you need not trade speed to get them. Under that objective:

- **Deepgram Nova-3 STT + ElevenLabs Flash v2.5 TTS: primary recommendation.** Deepgram’s STT ties Speechmatics for the best first-try number accuracy (70%) with zero permanent number losses and zero hallucination, lands the hard name “Mazin” once (fewer than AssemblyAI’s three, but unlike Speechmatics’ zero), and runs at roughly half Speechmatics’ latency (STT p50 0.374s vs. 0.777s, p95 0.465s). It is the best-balanced STT, with accuracy, reliability, and speed in one engine and no critical reliability weakness, though AssemblyAI captures the hard name more often. We pair it with ElevenLabs for synthesis because Deepgram’s own Aura TTS, not its STT, interrupted and cut off speech on the line; this stack also matches our production deployment.
- **AssemblyAI Universal-3 Pro: best for names / runner-up.** Best proper-noun accuracy (three exact “Mazin,” the only engine to get the brand acronym) and zero hallucination, but weaker first-try numbers (40%) with one permanent loss. The pick when proper-noun accuracy dominates.
- **Speechmatics Ursa (Enhanced): accuracy-focused alternative.** Ties Deepgram on first-try numbers (70%) with the cleanest digit auto-formatting, zero permanent losses, and zero hallucination, but runs $\sim 2\times$ slower (STT p50 0.777s) and never captured “Mazin” exactly, so its added latency buys no accuracy advantage over Deepgram. Choose it only where digit formatting matters most and latency is irrelevant.
- **ElevenLabs Scribe v2 Realtime: middle of the pack.** Zero hallucination, no permanent losses, low latency, but only 50% first-try numbers and one exact name, solid, not a category leader.
- **Cartesia Ink-Whisper: disqualified for telephony.** Fastest median, but hallucinates on phone audio (nine invented tokens) and has a broken 2s p95 tail. Concrete proof that speed $\neq$ reliability.

The decisive lesson generalizes beyond these five engines: evaluate the telephony *stack* on reliability under PSTN conditions, hallucination, permanent entity loss, and caller cut-offs, not on speed or on a single accuracy headline. Optimize for accuracy and reliability, but do not assume you must pay latency to get them, the best-balanced STT here is also one of the fastest, and the cut-off behavior that does break telephony lived in the TTS, fixed by pairing Deepgram STT with ElevenLabs TTS.

Limitations

This study is deliberately small and we state its limits plainly. It uses a *single speaker* and roughly ten calls per engine ($n \approx 10$), so per-engine differences of about one call are within noise and the proper-noun results rest on few hard tokens. We report *no* ground-truth WER; entity accuracy is measured through an **AI read-back proxy** that necessarily mixes STT and LLM behavior, so a read-back error is not always purely an STT error. Two “never” counts are judgment-dependent rather than mechanically reproducible: the AssemblyAI loss is a call on which the caller gave a different number, and the Cartesia loss depends on a non-monotonic read-back transcript (both footnoted in Section 3.2). The Aura-TTS interruption finding is *qualitative and observational*, a field observation that motivated our TTS choice; this benchmark held TTS constant at ElevenLabs and did not measure Aura, so we did not instrument a TTS interruption rate, and quantifying it is future work. The total-latency figures are sums of medians, not measured end-to-end percentiles, and exclude turn-detection wait and network time. All engines were run *unseeded*; seeding would likely shift the name and brand results. These caveats narrow the precision of the rankings but not their direction: the reliability separations, hallucination, permanent number loss, and the latency tail, are large enough to be decisive at this sample size.

References

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